D2.3 Water potential (HL)

NAME	DATE	TOTAL
		_____ / 20

About 20 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

5 MARKS

1 **DEFINE** water potential. [D2.3.8] [1]

2 **STATE** the unit of water potential. [D2.3.8] [1]

3 **STATE** the water potential of pure water. [D2.3.8] [1]

4 **STATE** the equation linking water potential, solute potential and pressure potential. [D2.3.10] [1]

5 **STATE** whether pressure potential is positive or negative in xylem sap under tension. [D2.3.10] [1]

Section B: Understanding

7 MARKS

6 **EXPLAIN** why dissolving a solute in water lowers its water potential. [D2.3.8] [2]

7 **EXPLAIN** the changes in solute potential and pressure potential when a flaccid plant cell is placed in pure water. [D2.3.11] [3]

8 **DEDUCE** whether a cell with a solute potential of −800 kPa and a pressure potential of +200 kPa gains or loses water in a solution of −400 kPa. [D2.3.9] [2]

Section C: Working with evidence

8 MARKS

All data in this section are invented for practice. The patterns are realistic.

9 [D2.3.11]

Potato cylinders were blotted, weighed, left in sucrose solutions for 24 hours, then blotted and reweighed.

SUCROSE / MOL DM^{-3}	0	0.2	0.3	0.4	0.6	0.8
MEAN CHANGE IN MASS / %	+16	+7	+2	−3	−10	−17
WATER POTENTIAL OF SOLUTION / KPA	0	−540	−820	−1120	−1760	−2460

a EXPLAIN the change in mass in pure water. [2]

b ESTIMATE the sucrose concentration at which there would be no change in mass. [1]

c ESTIMATE the water potential of the potato tissue. Show your reasoning. [2]

d EXPLAIN why percentage change in mass is used instead of change in mass. [1]

e EXPLAIN why the cylinders were blotted before each weighing. [1]

f SUGGEST one change to make the estimate more precise. [1]